National University of Sciences and Technology (NUST)

NUST Balochistan Campus (NBC)

DS-311 Data Mining

BS-AI 2024F, Section A · Fall 2026

Semester Project Proposal

AI Job-Market Skill Miner

Which skills go together in data and AI jobs, what job families exist, and what should a BS-AI student learn next?

Project P08 · **LinkedIn Job Postings 2023–2024**

Team Member	CMS ID	Role
Ahtisham Ul Hassan	529108	Team lead; salary-band classification
Quratulain Afzal	524410	Skill association rules
Rija Faisal	509132	Job-family clustering

Instructor: Dr. Zirak Khan

Lab Instructor: Syed Daniyal Asghar

Submission: Week 6 Lab, 14 October 2026

Repository: github.com/AHTISHAM327/ai-job-market-skill-miner

Data: LinkedIn Job Postings (2023–2024), A. Koneru et al., Kaggle, v13, CC BY-SA 4.0 • 1.3M LinkedIn Jobs & Skills (2024), asaniczka, Kaggle, ODC-By 1.0

Shared processed data keeps these licenses. AI use: drafted with Claude (Anthropic); all figures come from our Colab run and were checked by the team.

Problem

Students preparing for data and AI careers cannot easily see which skills employers ask for together, which kinds of roles exist, what to learn next, or how skills and experience relate to pay. Job ads hold this information only as unstructured text, so we mine real postings for evidence.

Data

LinkedIn Job Postings (2023–2024) on Kaggle by A. Koneru et al., version 13 (556.5 MB, 11 CSV files), license CC BY-SA 4.0. It is loaded in Colab from this pinned version with `kagglehub`, so all members use identical data. First checks:

`postings.csv`	123,849 postings, 31 columns; no duplicate job IDs; description missing in 7 rows
Time and place	First listed December 2023 to April 2024; about 95% of postings are in the United States
Pay disclosed	36,073 postings (29.1%): 20,628 yearly, 14,741 hourly, the rest monthly or weekly
Skills	Only 35 broad skill categories, and the `skills_desc` field is empty in 98% of postings
Data and AI subset	Chosen by job-title keywords: 1,772 postings; 583 disclose pay and 279 are entry level (preliminary)

For skills we also use the guide's alternative, *1.3M LinkedIn Jobs & Skills (2024)* (asaniczka, ODC-By 1.0), sampled to its data and AI postings: it lists extracted skills per job but has no salary data and only mid-senior and associate levels, so salary and entry-level analysis stay on the main dataset.

Research questions and methods

	Question	Owner	Method	How we judge it
Q1	Which skills appear together in data and AI postings, and which are most common?	Quratulain Afzal	Skill lists per job (from descriptions and the alternative skills dataset); FP-Growth association rules	Support, confidence and lift; hand check of extracted skills on sampled postings
Q2	What job families exist among data and AI postings?	Rija Faisal	TF-IDF on cleaned titles and descriptions, truncated SVD; k-means vs. hierarchical clustering	Silhouette score to choose *k* and compare both methods; clusters named by their top terms
Q3	What should a BS-AI student learn next, and what salary band can they expect?	Ahtisham Ul Hassan	Missing skills from the rules and job families; annualized pay, outliers removed, equal-frequency bands; random forest vs. decision tree	Accuracy and macro-F1 against a majority-class baseline on a stratified 5-fold CV; feature importance explains the drivers

Each question reports entry-level postings separately, as these roles matter most to students. In Q3, salary fields and money amounts in the text are excluded to prevent leakage. A first count on the 1,772 postings supports Q1: Python appears in 59% of descriptions, SQL in 50% and machine learning in 39%.

Plan

14 Oct • M0	This proposal, the dataset loaded in Colab with the pinned loader, and the GitHub repository with all members
21 Oct • M1	Data-quality report and EDA, including a hypothesis test
28 Oct • M2	Tables integrated and transformed; first baseline model and first association rules
18 Nov • M3	Core complete; clustering running; web-app skeleton with one working page
2 Dec • M4	Full evaluation (cross-validation, leakage check); app deployed at a public link
21–23 Dec	Report and repository submitted; demo, presentation and individual viva

All members work from one pinned data version, shared cleaning rules and numbered processed versions, recorded in a data manifest and GitHub.

Limitations

- U.S. LinkedIn postings from late 2023 to early 2024: results may not hold for Pakistan or later years, and the report discusses how far they transfer.
- Salary results cover only postings that disclose pay (29%; 583 data and AI postings); pay is annualized and extreme values (0 to $535.6 million a year in the full data) are cleaned first.
- Skill extraction can miss synonyms or confuse short names such as R or Go, and title keywords can include or miss roles; both are hand-checked on samples.